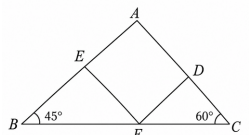


1. Dense Trace Initialization



**EF // AC and DF // AB.
Determine the measure of angle EFD.**

Verbose Chain-of-Thought

Okay, let me try to figure out the measure of angle EFD in this triangle setup. So, we have triangle ABC with angles at B and C given as 45 degrees and 60 degrees respectively. First, I should find the measure of angle A. Since the sum of angles in a triangle is 180 degrees, angle A must be $180 - 45 - 60 = 75$ degrees. So angle A is 75°. Now, there's a smaller triangle DEF inside ABC. Points D, E, F are on BC, CA, AB respectively. EF is parallel to AC, and DF is parallel to AB. Hmm, parallel lines mean that there might be similar triangles involved here. Let me visualize this. Since EF is parallel to AC, triangle AEF must be similar to triangle ABC... [omit 6206 tokens]

Raw Long-CoT Data



Visual

We have triangle ABC with angles at B and C given as 45 degrees and 60 degrees... [omit 568 tokens]

Think

First, I should find the measure of angle A. Since the sum of angles in a triangle is 180... [omit 424 tokens]

Answer

So the measure of angle EFD is 75°

Verbose (DROP)

Okay, let me try to figure out the measure of angle EFD in this triangle setup... [omit 5414 tokens]

Decompose into Three Parts



Visual

Triangle ABC; B=45°, C=60°. Segments: EF // AC, DF // AB. Need $\angle EFD$.

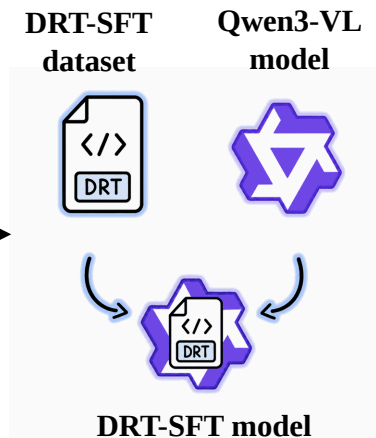
Think

[Priors] Sum of triangle angles=180°. Angle between parallel lines equals angle between their originals. → Compute A: $180^\circ - 45^\circ - 60^\circ = 75^\circ$. At F: DF // AB, EF // AC → $\angle EFD$ (angle between DF and EF) = angle between AB and AC = $\angle A = 75^\circ$.

Answer

$\angle EFD = 75^\circ$

Compress to Dense Trace



Supervised Fine-tuning

2. Trace-Grounded Reinforcement Learning

